

APK 4720 / APK 6725

AI for Movement Sciences

Introduction to Machine Learning

Introduction to Machine Learning

In its core, machine learning techniques allow us to build mathematical models to help us understand data.

These models have tunable (often called weights and biases) which are iteratively adjusted to minimize a loss function during the training process using data. This process allows us to **learn** from the data.

Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data.

Introduction to Machine Learning

Machine learning can be categorized in different ways. At the most fundamental level, it can be classified based of the type of data under analysis

- Supervised Learning involves modeling the relationship between measured features of data and some label associated with the data. Supervised Learning is further subdivided into classification tasks and regression tasks: in classification, the labels are discrete categories, while in regression, the labels are continuous quantities.
- Unsupervised Learning involves modeling the features of a dataset without reference to any label. Unsupervised Learning is further subdivided into tasks such as clustering and dimensionality reduction.

Introduction to Machine Learning

Classification – Predicting discrete labels

In a classification problem, there are typically multiple features (or measures) and each feature set is associated with a label (or class).

In this example, we have two features for each data point, and a label, represented by the color of each point.

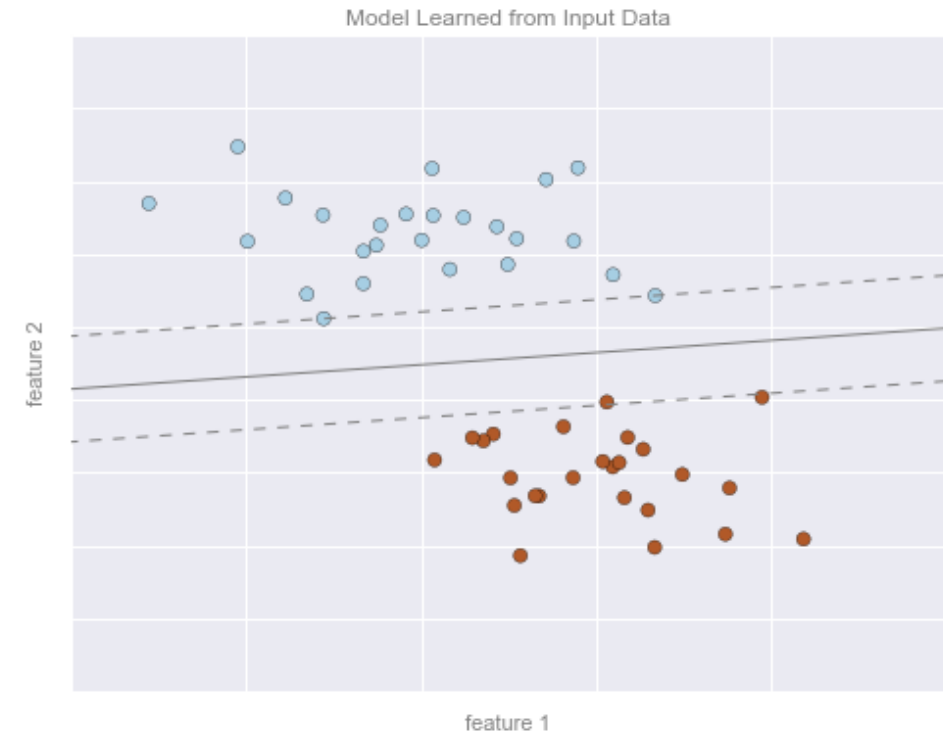


Introduction to Machine Learning

Classification – Predicting discrete labels

The objective of the classification problem is to define a boundary in the feature space that can separate the classes. This boundary can be a line or a more complex shape.

Classification problems usually have more than two features, so the separation in the feature space might not be intuitive.



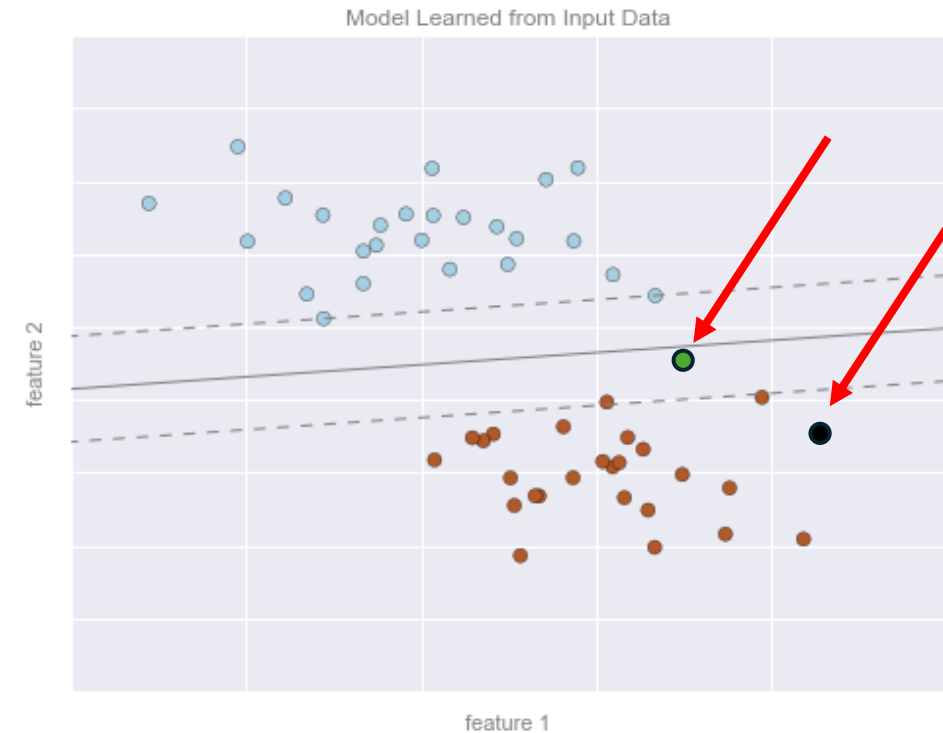
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Classification – Predicting discrete labels

Once the boundary is defined, we can use it to classify new data that does not have a label.

What is the label of the black dot ?

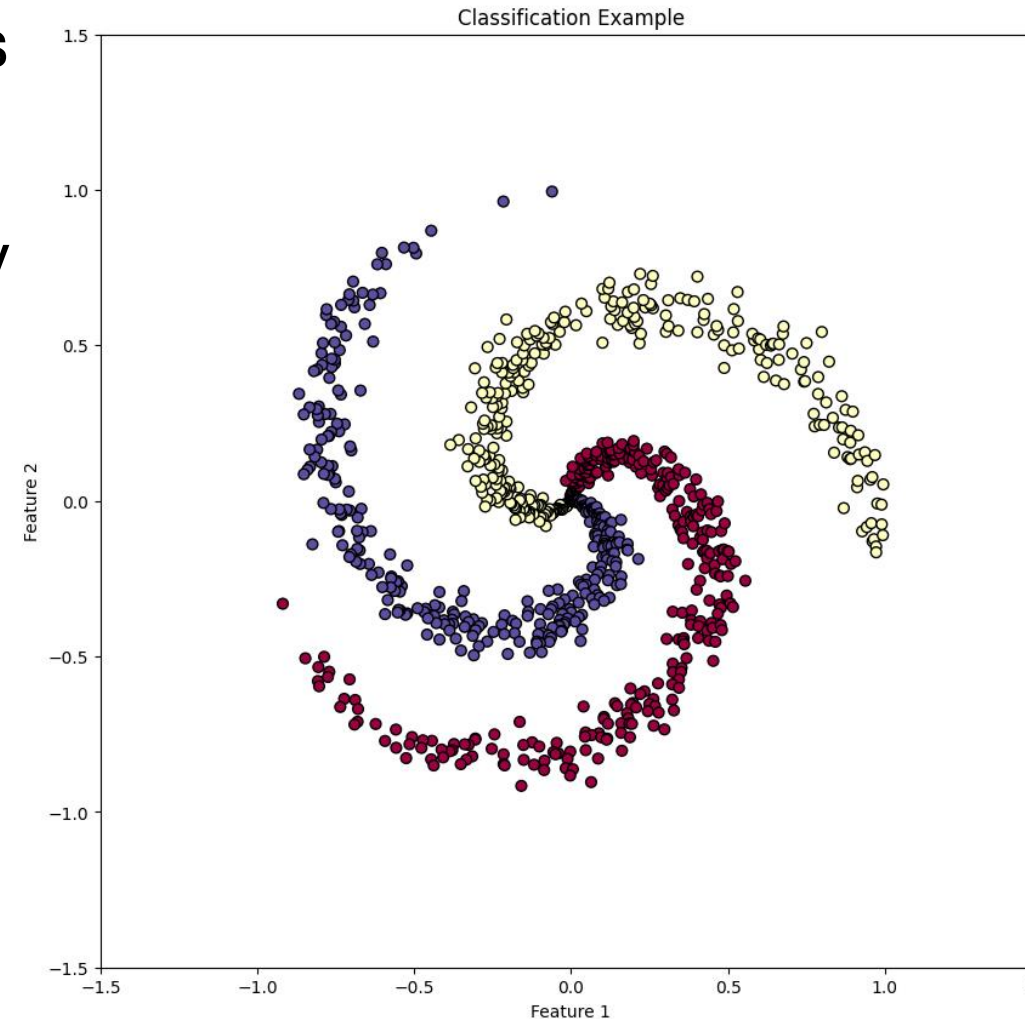
What is the label of the green dot ?



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Classification – Predicting discrete labels

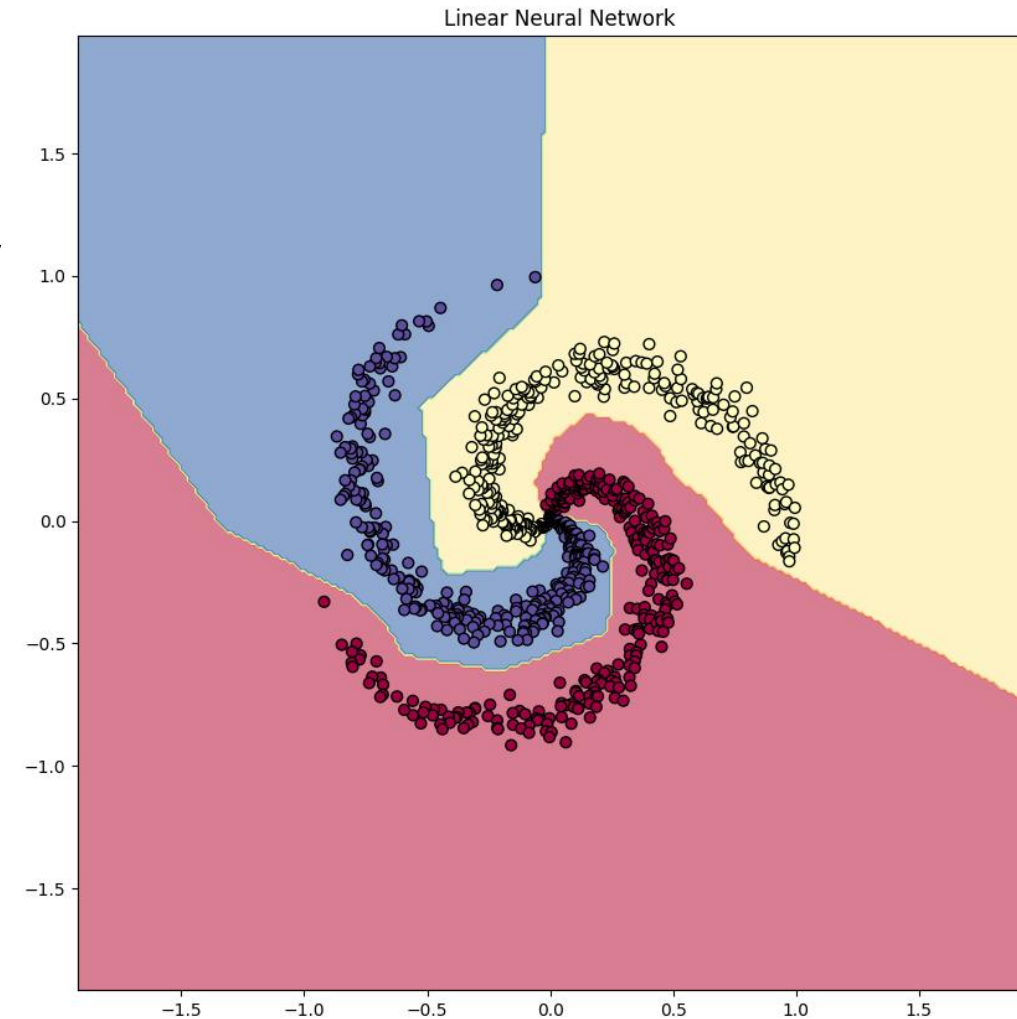
Not all classification problems are linearly separable, we will learn methods that can learn non-linear boundaries to separate the data.



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Classification – Predicting discrete labels

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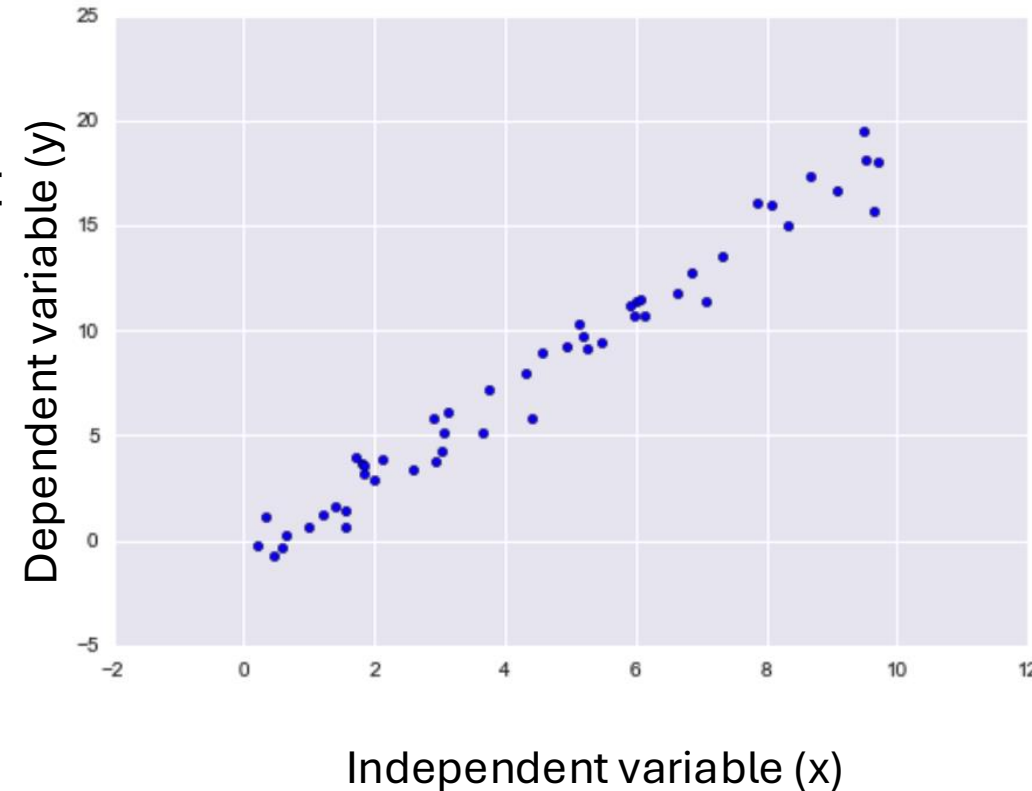


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Regression – Predicting continuous labels

In a regression problem, there are typically one or multiple features (called independent variable(s)) and each feature set is associated with an output (or dependent variable).

In this example, we have one independent variable (plotted in the x-axis) and one dependent variable (plotted in the y-axis)



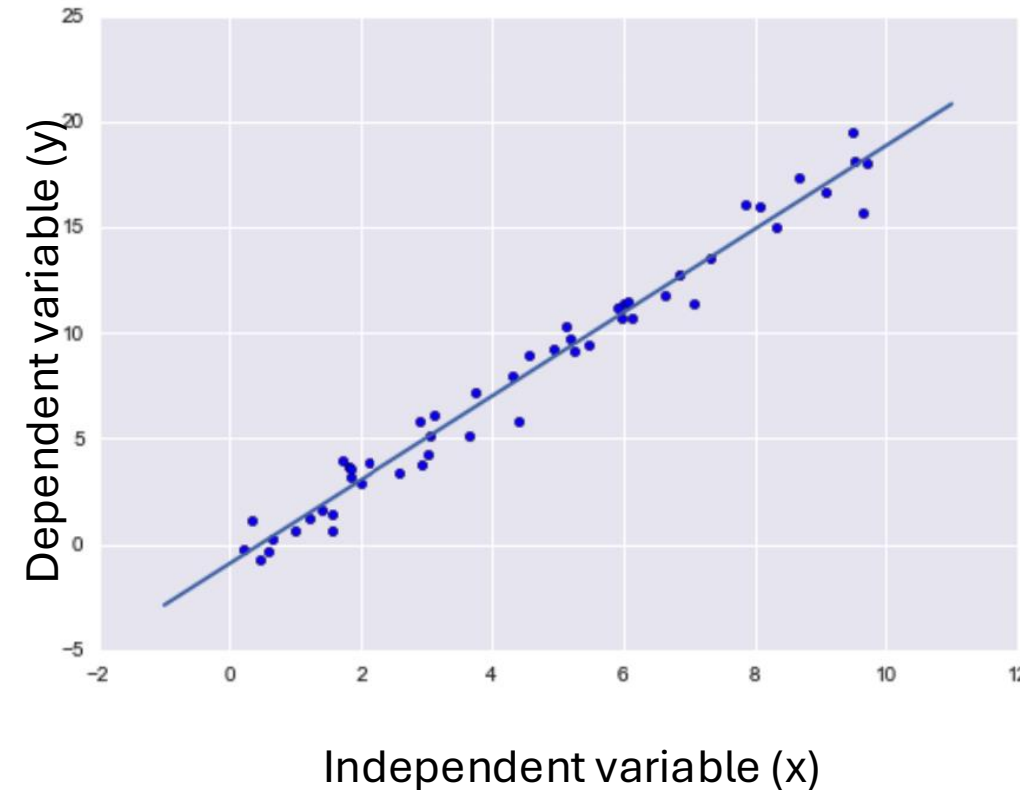
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Regression – Predicting continuous labels

In this example, the dependent (y) and independent (x) variables are related

$$y = \omega_0 + \omega_1 x + \epsilon$$

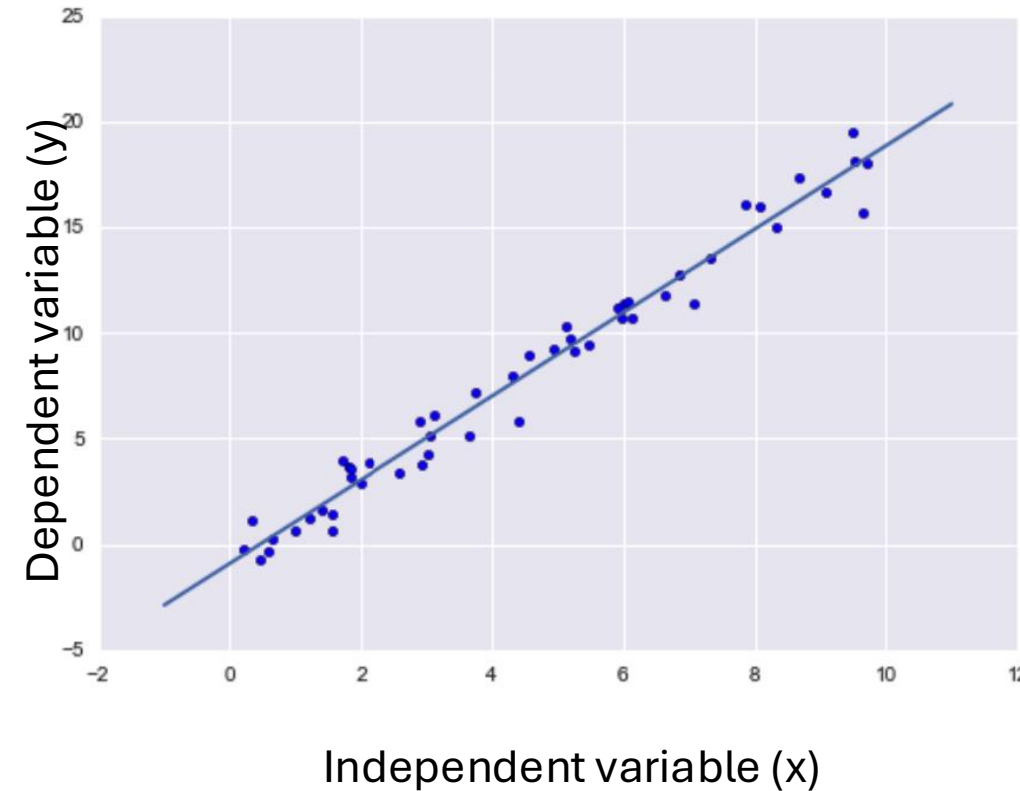
Where ω_i are the model weights, and ϵ is an error term.



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Regression – Predicting continuous labels

The objective of the regression problem is to identify the weights (ω_0 and ω_1) that minimize the differences between the measured data and the model's predictions.



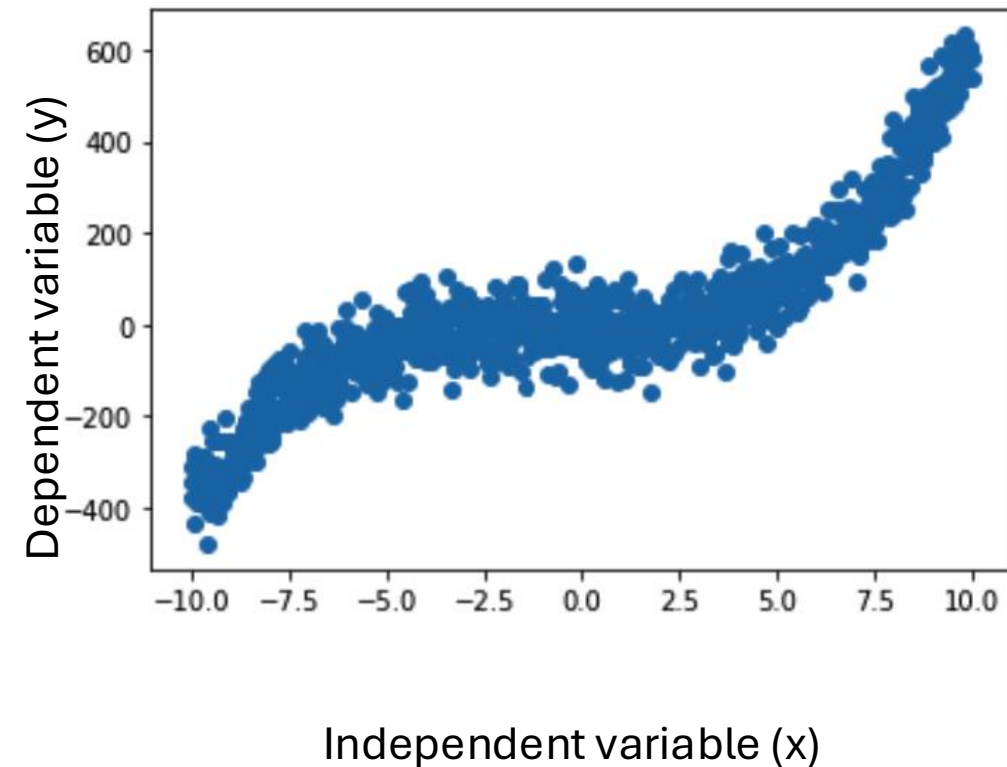
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Regression – Predicting continuous labels

Note that the relation between the dependent and independent variables does not need to be linear.

Similarly, a regression problem can have multiple independent variables

$$y = \omega_0 + \omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + \dots + \epsilon$$

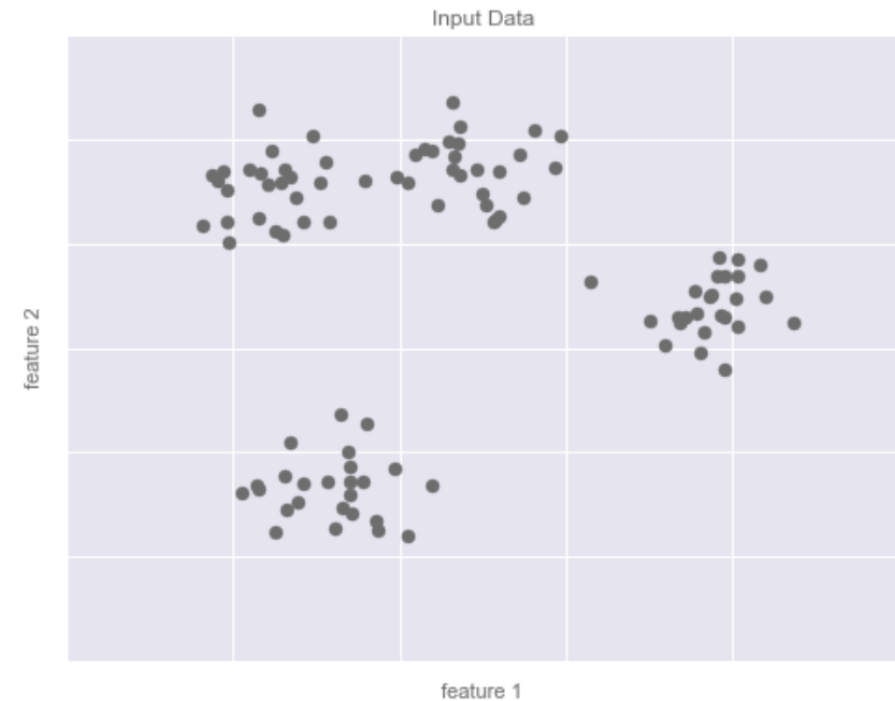


Introduction to Machine Learning

Clustering – Inferring labels from data

Clustering is a common unsupervised learning problem. In a clustering problem, the features can typically be separated into different groups.

A clustering algorithm will use the structure of the data to determine how many groups are present and the class for each measurement.



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Clustering – Inferring labels from data

In this example, the data can be separated into four groups.

Once the groups are identified, it is possible to assign new data to a cluster.

What is the cluster of the black dot ?

What is the cluster of the orange dot ?

